

AARON FOOTE

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EDUCATION

UC Berkeley , Berkeley, CA	September 2026 - Present
M.A. in Biostatistics	
Wesleyan University , Middletown, CT	September 2020 - May 2025
M.A. in Computer Science	GPA: 4.22/4.00
B.A. with High Honors in Computer Science, Minor in Data Analysis	GPA: 3.87/4.00

RESEARCH EXPERIENCE

NIOSH ERC Trainee	September 2026 - Present
<i>UC Berkeley Center for Occupational and Environmental Health</i>	<i>Berkeley, CA</i>

- Applying flexible ML causal inference to evaluate risk of occupational exposure in longitudinal cohorts
- Developing causal inference methods to improve the power and robustness of TMLE in longitudinal applications

Research Associate	June 2025 - August 2026
<i>CMC Foundation for Change</i>	<i>Palo Alto, CA</i>

- Coordinated survey data collection with clinicians and community leaders
- Overhauled database with hundreds of surveys on substance abuse treatment trainings with Python and SPSS
- Developed data visualizations and reports for internal decisions and external fundraising/presentation
- Mentored small team of research assistants on data cleaning, data management, data analysis, and report writing processes

Graduate Research Assistant – Interpretable ML	June 2024 - August 2025
<i>Wesleyan University</i>	<i>Middletown, CT</i>

- Developed a hypothesis test under minimal assumptions to evaluate feature importance methods for bias
- Validated the method through extensive simulation and application to real datasets in Python and R
- Integrated dimension reduction techniques to extend the test to higher dimension

Graduate Research Assistant – Interpretable ML	June 2024 - August 2025
<i>Wesleyan University</i>	<i>Middletown, CT</i>

- Developed a hypothesis test under minimal assumptions to evaluate feature importance methods for bias
- Validated the method through extensive simulation and application to real datasets in Python and R
- Integrated dimension reduction techniques to extend the test to higher dimension

Research Assistant – Sports Analytics	January 2024 - June 2024
<i>Wesleyan University</i>	<i>Middletown, CT</i>

- Produced an intuitive evaluation tool for NFL defenders using spatiotemporal player tracking data
- Leveraged ensemble ML algorithms to estimate player value and assign credit to defenders
- Designed and deployed an NCAA March Madness prediction model using Cox proportional hazards and bootstrapping
- Explored time-varying Cox model and random/deterministic bracket completion through extensive simulation

TEACHING EXPERIENCE

Substitute Teacher/Paraeducator	September 2025 - July 2026
<i>Swing Education</i>	<i>Palo Alto, CA</i>

- Foster a positive and inclusive learning environment that supports diverse student needs
- Deliver engaging lessons across multiple grade levels and subject areas
- Manage classroom behavior effectively by applying school policies
- Provide individualized academic assistance aligned with each student's IEP goals and accommodations

Head Quantitative Analysis Tutor

Wesleyan University

April 2023 - May 2025

Middletown, CT

- Held 10+ hours of weekly office hours open to the entire university, assisting with coursework, projects and research
- Provided one-on-one statistical consulting for undergraduate and graduate thesis projects
- Guided students in managing and analyzing data, employing and interpreting statistical techniques, and visualizing their work with R, STATA, Python, SPSS, SAS, and Excel
- Developed a schedule each semester, lead weekly meetings, and consult with professors

Teaching Assistant

Wesleyan University

September 2022 - December 2024

Middletown, CT

- Led weekly lab sections for 15+ students, explaining complex concepts through hands-on activities
- Graded 50+ assignments weekly on Moodle, providing detailed feedback to improve student understanding
- Courses: Algorithms, Theory of Computation, Survival Analysis, Applied Data Analysis

RESEARCH PRESENTATIONS

Foote, A. and Krizanc, D. (2025) "Nonogram: Complexity of Inference and Phase Transition Behavior." *arXiv Pre-Print* doi:10.48550/arXiv.2507.07283

Foote, A. and Krizanc, D. (2025) "TRIP: A Nonparametric Test to Diagnose Biased Feature Importance Scores." *IJCAI Workshop on Explainable Artificial Intelligence* (Oral Presentation) doi:10.48550/arXiv.2507.07276

Foote, A. and Matthews, G. (2024). "Continuous Within-Play Estimation of Run-Stopping Contribution in American Football." *Cascadia Symposium on Statistics in Sports* (Poster).

Foote, A., Gooyabadi, M. and Addleman, N. (2023). "Factors in Learning Dynamics Influencing Relative Strengths of Strategies in Poker Simulation." *Games*. doi:10.3390/g14060073.

Foote, A., Oleinikov, P. (2021). "Efficiently Extracting Text from Political Ad Videos." *CIS Summer Research Symposium*. (Poster)

AWARDS

Maureen Snow Prize (July 2023): Runner-up among 200 students, given for most effective communication of research to a broad audience

Baker '64 Collabria Fellowship in Data Analysis (April 2023) – \$5,000

Best Data Story Award (April 2023): Awarded at Northeast Division ASA DataFest Competition

QAC Apprentice Grant (April 2021) – \$5,400